Vladislav Larionov

Quantitative research · MS in Financial Mathematics · Python
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SUMMARY

Second-year MS student in Financial Mathematics and Financial Technologies at Sirius University. Research on volatility modeling for crypto options: processing large volumes of high-frequency market data, calibrating volatility models and analyzing their dynamics. Two-time prize-winner of the All-Russian Chemistry Olympiad. Research experience since 2021, data preparation for ML models, one Q1 publication.

RESEARCH PROJECTS

Implied volatility surfaces for BTC and ETH options — MS research, supervisor R.V. Grinis (MIPT) *2025 — present*

- Built a Python pipeline of Deribit options tick data (Nov 2024 – Apr 2025): nanosecond timestamps, OTM filtering, log-moneyness axes; rewrote the hot loops in Numba (@njit).
- Calibrated SABR (separate ATM branch) and SVI to daily smiles; checked the surfaces for arbitrage.
- Built time series of model parameters and studied the distributions of their log-returns: Jarque–Bera and Shapiro–Wilk tests reject normality.
- Stack: pandas, NumPy, Numba, SciPy, Matplotlib, Parquet.

Topic modeling and multi-label text clustering — team project, GazpromBankTech course *Dec 2025*

- Scraped user reviews from Banki.ru with Python (BeautifulSoup).
- Used an LLM to extract the topics of each review and the sentiment toward each topic.

Reproduction of "Differential Machine Learning" (Huge & Savine, 2020) — team project, Yandex SHAD course *Oct 2026*

- Goal: reproduce the paper on its public examples: a 7-asset basket under the Bachelier model and a 4-asset worst-of autocallable under local volatility.
- Networks are trained on differential labels (pathwise derivatives via AAD) with a loss that combines price error and λ-weighted error in the Greeks.
- Implemented differential regression as a Python package (NumPy, SciPy, scikit-learn) with Jupyter notebooks; twin network (PyTorch) and differential PCA in progress. Code

EDUCATION

Sirius University — MS, Financial Mathematics and Financial Technologies *2025 — present*

- Topics: stochastic calculus, algorithmic trading and market making.

Lomonosov Moscow State University — Faculty of Chemistry *2017 — 2023*

EXPERIENCE

Sibur PolyLab — Chief Specialist *Jan 2025 — Aug 2025*

- Collected, cleaned and analyzed IR-spectra data for training and validating ML models together with the SBER AIRI team.
- Tested and configured the LIMS, gathered requirements for new features, wrote and rolled out SOPs.

Sirius University — Junior Researcher, Biomaterials *Oct 2023 — Apr 2025*

A.N. Belozersky Institute of Physico-Chemical Biology, MSU — Protein Chemistry Department *Feb 2021 — Jun 2023*

ACHIEVEMENTS, PUBLICATION, EVENTS

- 7th International Summer School of Financial Mathematics, Kaluga (Aug 2026).
- Hackathon 4.0, Sirius — 3rd place (May 2026).
- MOEX & FX Forum (May 2026).
- Gevorgyan K.P., Larionov V.I., Akopyan A.V., Anisimov A.V., Domashkina P.D. Room Temperature Ultrafast Oxidative Desulfurization with Sodium Hypochlorite in the Presence of Silica-Supported Catalysts. ACS Omega 2024, 9, 13831–13839 ***(Q1, IF 4.1)***
- Conferences: BRICS Congress on Ecology and Climate Change (2024); III–V Congresses of Young Scientists (2023–2025).
- All-Russian Chemistry Olympiad for schoolchildren — two-time prize-winner (2016, 2017).
- Grant of the President of the Russian Federation (2017–2023).

SKILLS

Programming: Python (NumPy, pandas, SciPy, Numba, scikit-learn, Matplotlib), Jupyter, Parquet, LaTeX
Quant: option pricing and volatility models (SABR, SVI), stochastic calculus, hypothesis testing, numerical optimization
Languages: Russian (native), English (B2)
Interests: ultimate frisbee (team sport), cycling, diving (PADI OWD)